AI-ML: Study Pattern Analyzer

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Project Title: Study Pattern Analyzer

◆ **Overview of the Project:**

This application predicts a student's performance category based on their daily habits, specifically *study hours, sleep duration, and level of distractions. It utilizes a **Decision Tree Classifier* model trained on sample behavioral data to provide personalized feedback and suggestions for improvement.

◆ **Features:**

- *Behavioral Prediction:* Uses a Decision Tree algorithm to categorize performance into "Good," "Average," or "Poor."
- *Real-time Analysis:* Takes direct user input for daily metrics to provide an instant academic outlook.
- *Personalized Suggestions:* Evaluates specific input thresholds to offer actionable advice (e.g., increasing study time or reducing distractions).
- *Error Handling:* Utilizes integer casting to ensure user inputs are processed correctly for the ML model.

◆ **Technologies/Tools Used:**

- *Programming Language*
    - *Python 3.x:* Used to implement the logic, handle user inputs, and manage data structures.
- *Libraries*
    - *Scikit-learn:* Used to implement the DecisionTreeClassifier for the AI-based performance prediction.
- *Runtime Environment*
    - *Google Colab / Jupyter Notebook:* Executes the code blocks and displays the interactive terminal interface.

◆ **Steps to Install & Run the Project:**

Instructions for Testing:

1. Open *Google Colab* or *VS Code* with the Python extension installed.
2. Ensure you have the required library installed by running: pip install scikit-learn
3. Copy the code into a .py file or a notebook cell.
4. *How to Run the Program:*
    - Click the *Play/Run* button in your IDE or Colab cell.

    - Follow the on-screen prompts in the terminal/output window.

◆ **Expected Program Flow:**

- System initializes the sample dataset (Study hours, sleep, distractions) and trains the *Decision Tree* model.
- User enters their *Name*.
- User enters *Study Hours*.
- User enters *Sleep Hours*.
- User enters *Distractions* (on a scale).

Program Displays:

- A performance prediction: *"Looks like you will do good," "Maybe average performance,"* or *"Performance might be low."*
- Specific suggestions based on the user's habits (e.g., "sleep a bit more").